

7	(a) (i) lowest frequency of e.m. radiation giving rise to emission of electrons (from the surface)	M1	A1	[2]
	(ii) $E = hf$ threshold frequency $= (9.0 \times 10^{-19}) / (6.63 \times 10^{-34})$ $= 1.4 \times 10^{15} \text{ Hz}$	C1	A1	[2]
	(b) <i>either</i> $300 \text{ nm} \equiv 10 \times 10^{15} \text{ Hz}$ (and $600 \text{ nm} \equiv 5.0 \times 10^{14} \text{ Hz}$) <i>or</i> $300 \text{ nm} \equiv 6.6 \times 10^{-19} \text{ J}$ (and $600 \text{ nm} \equiv 3.3 \times 10^{-19} \text{ J}$) <i>or</i> zinc $\lambda_0 = 340 \text{ nm}$, platinum $\lambda_0 = 220 \text{ nm}$ (and sodium $\lambda_0 = 520 \text{ nm}$) emission from sodium <u>and</u> zinc	M1	A1	[2]
	(c) each photon has larger energy fewer photons per unit time fewer electrons emitted per unit time	M1	M1	A1
				[3]

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